

Dishari: A Novel Gesture-Based Educational Application for Specially Challenged People

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Abstract. This research explores the integration of advanced technologies, particularly hand gesture detection, and Generative AI, to create an inclusive educational tool called "Dishari" tailored for individuals with hearing and speech impairments. Traditional search engines often fail to address the specific needs of this speech and visually-impaired demographics, leading to a gap in accessibility and inclusivity. Leveraging machine learning algorithms and computer vision technologies, Dishari interprets hand gestures captured via web cameras, translating them into sign language expressions and helping the users search for a particular keyword through the search engine. Furthermore, the incorporation of Generative AI enhances the search functionality, enabling users to input queries through both text and hand gestures. Through a comprehensive literature survey, we highlight the advancements in hand gesture recognition systems and the transformative potential of Generative AI in image and text-based search. Dishari marks a significant milestone in bridging the communication gap and fostering inclusivity by empowering individuals with hearing and speech impairments to navigate the digital landscape effectively.

Keywords: Hand Gesture Recognition · Generative AI · Inclusive Education · Machine Learning · Computer Vision · MediaPipe · Accessibility · Sign Language · Educational Technology

1 Introduction

The usage of computers and IT has evolved exponentially in recent years [1]. From Banking sectors to healthcare; and educational institutes to defense, computers have taken the entire control [2]. In education, computers have given access to unlimited global resources with IoT to the users thereby enhancing the knowledge of the kids thoroughly. Unfortunately, little has been done for people with special abilities.

One of the significant learning tools is to accomplish an effective search online. Over the years, the landscape of text-based search has undergone significant transformations. Early efforts in information retrieval, dating back to 1948 [3],

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laid the foundation for organizing and accessing data stored in databases. Traditional search engines utilized keyword matching techniques, often resulting in irrelevant search queries due to discrepancies in user-inputted keywords and web page content [4]. To address these limitations, researchers explored diverse approaches to improve search results. Multi-layer indexing, introduced to deep crawl methods, is aimed at enhancing the effectiveness of data retrieval [5]. Various methodologies such as metadata-based [6] and semantic search [7] were proposed, each with its set of advantages and constraints.

In the realm of educational image processing engines, the integration of machine learning gained prominence. Pre-trained models were employed to predict search results, offering a more refined and accurate outcome [4]. However, challenges persisted, especially in cases where irrelevant data was uploaded by users, contributing to the need for advanced techniques [4]. Drawing from this historical context, our paper builds on prior research efforts, incorporating advancements in machine learning and Generative AI. We delve into the synergy of hand gesture detection and search functionality, providing an inclusive solution for individuals with hearing and speech impairments. Our approach not only addresses the limitations of traditional search methods but also empowers users with a more intuitive and interactive means of navigating the vast landscape of information on the internet.

We categorize the motivations behind conducting this research in the following subsections (see Sect. 1.1, 1.2 and 1.3). Later in Sect. 1.4 we discuss the novelty of our education tool Dishari.

1.1 Hand Gesture Recognition

Hand gestures, a fundamental form of non-verbal communication, have a rich history deeply intertwined with human expression and cultural diversity [8]. From ancient civilizations to modern societies, gestures have conveyed emotions, commands, and meanings across linguistic barriers. With the advent of the internet, hand gestures have gained a new dimension, playing a pivotal role in human-computer interaction [9]. The integration of gesture-based interfaces, such as touchscreens and motion-sensing devices, has transformed the way we interact with technology. Moreover, the rise of social media platforms and video communication tools has facilitated the widespread use of hand gestures as a means of expression in the digital realm [10]. On the frontier of technological innovation, there is a growing interest in the development of systems capable of detecting and interpreting hand gestures, particularly in the context of sign language. This has profound implications for accessibility, enabling individuals with hearing and speech impairments to communicate effectively through virtual platforms. Advanced machine learning algorithms and computer vision technologies are being employed to create gesture recognition systems that can accurately interpret intricate movements and translate them into meaningful communication [11]. This intersection of hand gestures and the digital landscape not only enhances user experience but also fosters inclusively by breaking down communication barriers. As technology continues to evolve, the synergy between hand

gestures and the internet is likely to play an increasingly significant role in shaping the way we communicate, bridging the gap between physical and virtual realms, and fostering a more inclusive and expressive digital environment.

Our web application leverages advanced hand gesture recognition technology by integrating MediaPipe's Gesture Recognizer with machine learning algorithms. MediaPipe's Gesture Recognizer utilizes pre-packaged model bundles to detect hand landmarks and recognize gestures accurately [12]. For training, we used the American Sign Language (ASL) Alphabet Dataset from Kaggle [13], ensuring the model is robust and accurate in interpreting a wide range of gestures. By utilizing this dataset, we have tailored the gesture recognition capabilities specifically for ASL, enhancing the system's effectiveness in real-time communication [14]. The combination of MediaPipe and the ASL dataset allows our application to provide precise and reliable gesture recognition, facilitating seamless interaction for users.

In practical application, our system operates seamlessly, detecting and interpreting hand movements captured by cameras. As users express themselves through sign language, our technology translates their gestures into corresponding sign language expressions, facilitating inclusive communication. By harnessing the synergy of MediaPipe's pre-packaged model bundles, our application "Dishari" not only addresses the communication needs of the deaf and hard-of-hearing community but also demonstrates the transformative potential of technology in fostering inclusivity and accessibility especially for the specially-abled people [15].

1.2 Search Powered by Generative AI

Image and text-based search input powered by Generative AI has revolutionized the way users interact with digital content [16]. Leveraging advanced algorithms, Generative AI enables the generation of relevant textual descriptions or visually similar images based on user queries [17]. This technology finds applications across diverse domains, enhancing user experiences and efficiency [18]. In e-commerce, users can now effortlessly find products by describing them in natural language or uploading images, leading to more accurate search results and increased customer satisfaction [19]. In the realm of content creation, Generative AI facilitates the exploration of creative ideas by suggesting relevant images or generating textual content based on user inputs whether it is an image, voice, or text [20]. Furthermore, in healthcare, this technology assists in medical image analysis, allowing for more precise diagnosis and treatment planning [21]. Overall, the integration of Generative AI in image and text-based search inputs streamlines information retrieval processes and fosters innovation across industries, making it an indispensable tool in the modern digital landscape.

In our smart application "Dishari", we have incorporated Generative AI to make our application stronger, wider, and more convenient for the users(especially for people with special abilities).

1.3 Applications of Image Processing and Analysis Through Generative AI

In this subsection, we will discuss various applications of using Image Processing and Analysis through Generative AI step by step.

1.3.1 Education Enhancement

Sign language detection with hand gestures using MediaPipe's Gesture Recognizer, integrated with Image and text-based search input powered by Generative AI, can significantly enhance educational experiences for individuals with hearing and speech impairments [22]. By translating sign language gestures into text or images, Generative AI enables interactive communication with digital devices, facilitating a more inclusive and effective learning environment [23].

1.3.2 Healthcare Communication

Generative AI has valuable applications in the healthcare sector [24], where effective communication is crucial. Sign language detection enables healthcare providers to understand and respond to patients with hearing and speech impairments. Integrating this information into Generative AI-powered search inputs ensures an accurate and efficient exchange of medical information, improving the overall quality of healthcare communication for patients and doctors [25].

Dishari is a novel hand gesture-based application that has worked with Generative AI to interpret sign language into text and help specially-abled people communicate with each other and the rest of the world through the internet.

1.4 Contributions of Dishari

Dishari is a dedicated and novel application, especially designed for specially-abled people to connect themselves with the world. The primary notion of Dishari is to open a broad spectrum of educational tools to physically challenged people and make them capable of grasping knowledge from the Internet. The main novelty of this research work lies in connecting the world of hand gesture detection with that of generative AI. This is the first application where a physically challenged user can simply use their hand gestures to search for a keyword on the application.

Dishari marks a significant milestone in bridging the gap between human gestures and machine interaction. We discuss the step-by-step processing of our novel gesture-based educational application "Dishari" for specially challenged people as follows.

- Using MediaPipe's Gesture Recognizer, Dishari, a novel gesture-based application, captures hand gesture inputs and processes them to identify hand movements with remarkable accuracy. MediaPipe's advanced technology allows for real-time detection of hand landmarks and gesture recognition, ensuring precise interpretation of user gestures.

- The application integrates a gesture recognition model trained on the American Sign Language (ASL) Alphabet Dataset from Kaggle. This model, fine-tuned specifically for ASL gestures, enables the application to accurately recognize and interpret sign language gestures, providing precise extraction of gesture labels corresponding to the detected hand movements.
- Taking user experience to the next level, the application seamlessly integrates with a search bar, enhanced by Generative AI capabilities. The main novelty of this research lies in this phase. No other researchers earlier have connected gesture detection with generative AI.
- Upon recognizing a sign language gesture, the application automatically populates the search bar with relevant keywords, providing intuitive interaction and accessibility for users.

This innovative fusion of computer vision, machine learning, and Generative AI opens doors to a world where technology empowers communication and accessibility for all.

Next, we discuss the pre-existing research on Hand gesture recognition systems and Generative AI. We also mention how our project has combined both worlds and created a novel application for people with special abilities in Sect. 2.

2 Literature Survey

Hand gesture recognition systems have witnessed significant advancements over the years, driven by the increasing demand for intuitive human-computer interaction interfaces [26]. This section provides an overview of the existing literature on hand gesture recognition systems, highlighting key methodologies, challenges, and applications. Also, we focus on how our novel application "Dishari" is unique from most of the pre-existing research.

2.1 Background of Hand Gesture Recognition Systems

The roots of hand gesture recognition can be traced back to the early developments in computer vision and pattern recognition. Notably, Vicon, a motion capture system introduced in the 1980s, laid the groundwork for capturing and analyzing human movements [26]. However, it wasn't until the late 20th century that research in gesture recognition gained momentum with the emergence of robust algorithms and computational resources [27].

Several methodologies and techniques have been explored in hand gesture recognition, ranging from traditional computer vision approaches to modern deep learning methods. Early systems relied on feature-based approaches, extracting handcrafted features such as shape, texture, and motion to classify gestures [28]. However, these methods often suffer from limited robustness and scalability.

With the advent of deep learning, particularly Convolutional Neural Networks (CNNs), the landscape of hand gesture recognition underwent a paradigm shift. CNN demonstrated superior performance in learning hierarchical features from raw data, enabling more accurate and robust gesture recognition

[27]. Transfer learning techniques have also been leveraged to adapt pre-trained models for gesture recognition tasks, mitigating the need for extensive labeled datasets [29].

Despite significant progress, hand gesture recognition systems still face several challenges. One notable challenge is the variability in gesture appearance and articulation across different users and environments [30]. This variability poses a significant hurdle in designing robust and generalizable recognition algorithms [31]. Additionally, real-time processing requirements impose constraints on computational efficiency and latency, necessitating lightweight models and optimized algorithms [32].

Hand gesture recognition has found diverse applications across various domains. In Human-Computer Interaction (HCI), gesture-based interfaces offer intuitive and immersive interaction experiences, particularly in virtual and augmented reality environments [33]. In robotics, gesture recognition enables natural communication between humans and robots, enhancing collaboration and task execution [34].

We discuss our next focus "Generative AI" in the next subsection(see Sect. 2.2. We discuss how our novel educational application Dishari perfectly blends into the domain of the Gesture recognition system and generative AI in Sect. 2.3.

2.2 Generative AI in Image and Text-Based Search

Generative Artificial Intelligence (GAI) has revolutionized image and text-based search, empowering users with advanced retrieval capabilities and enhancing user experiences [16]. Generative AI has its roots in the field of deep learning and generative modeling, with seminal works such as Generative Adversarial Networks (GANs) and Variational Autoencoders (VAEs) paving the way for realistic data generation [35]. These models enable the synthesis of novel images, texts, and even multimedia content, revolutionizing the landscape of content creation and search.

In image-based search, Generative AI facilitates content-based image retrieval by generating visually similar images based on user queries [36]. Techniques such as Conditional GANs allow for the generation of images conditioned on specific attributes or characteristics, enabling fine-grained search and customization [37]. Similarly, in text-based search, Generative AI models such as Language Models (e.g., GPT series) generate textual descriptions or summaries based on user input, improving search relevance and comprehension [38].

Despite its transformative potential, Generative AI in search is not without challenges. Ethical considerations, such as the potential for bias and misinformation in generated content, require careful attention [39]. Furthermore, ensuring the robustness and reliability of generative models against adversarial attacks and data biases remains an ongoing research endeavor [40]. Future directions in this field include the development of more interpretable and controllable generative models, as well as addressing scalability and efficiency concerns for real-world deployment [41].

We discuss how we incorporate generative AI in our application "Dishari" in Sect. 2.3.

2.3 Dishari and Its Implementation

In comparison to existing hand gesture recognition systems, Dishari offers several unique features and advantages tailored specifically for individuals with hearing and speech impairments. Firstly, Dishari incorporates advanced machine learning algorithms, including CNNs, trained on a diverse dataset of sign language gestures, ensuring robustness and accuracy in interpretation. This specialized training enables Dishari to recognize a wide range of sign language gestures with high precision, making it suitable for practical use in real-world scenarios.

Moreover, Dishari leverages Generative AI technology to enhance the search functionality, enabling users to input queries through both text and hand gestures. By combining hand gesture recognition with Generative AI-powered search, Dishari provides a seamless and intuitive interaction experience for users, enabling them to access information effortlessly.

Furthermore, Dishari is designed with inclusivity in mind, aiming to bridge the communication gap between individuals with hearing and speech impairments and the rest of the world. By providing a platform for interactive communication through sign language interpretation and text-based search, Dishari empowers users to express themselves effectively and access information independently.

In summary, hand gesture recognition has seen significant advancements in recent years, with a shift towards data-driven approaches and the integration of machine learning techniques. Dishari represents a novel approach that combines hand gesture recognition with Generative AI-powered search, offering unique advantages for individuals with hearing and speech impairments.

3 An Insight into Dishari

In this section, we are going to discuss the requirements for using Dishari; the underlying methodology, and the architectural framework for developing it.

3.1 Hardware Requirement

To use Dishari, we require the three fundamental components mentioned below.

- A computer with stable internet connection.
- A web browser (latest version of Google Chrome or Safari or Microsoft Edge or Firefox).
- A webcam (which captures the hand gestures).

Figure 1 depicts the basic prototype of Dishari. Dishari enables users to interact with their computer webcam through hand gestures. These gestures are then processed by a machine learning (ML) model(MediaPipe gesture recognition)

embedded within the application. If a recognized gesture is detected, the underlying object detection API (an integration of deep learning algorithms like CNN) analyzes it and provides an output. This output typically includes hand landmarks with colored lines of detected hands along with their name and accuracy. Following this recognition process, the detected gesture information is fed as input for a Generative AI system. Users can initiate searches for desired keywords by pressing the enter button once they are ready. This integration of hand gesture recognition and Generative AI search functionality forms a pivotal aspect of the Dishari, contributing to its goal of providing an inclusive solution for individuals with hearing and speech impairments (refer Fig. 1).

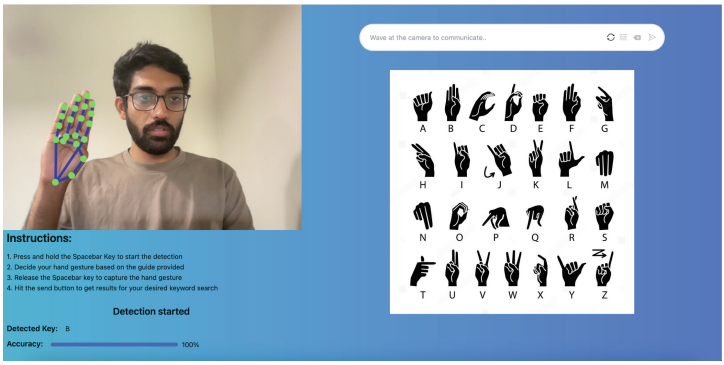


Fig. 1. Dishari prototype

Now we will discuss the underlying methodology implemented in the Dishari in Sect. 3.2.

3.2 Methodology

In this section, we describe the underlying methodology implemented in Dishari.

We describe the entire algorithm in this research to develop Dishari, a hand gesture-based educational application. We describe every step diagrammatically in Fig. 2.

- The first step is to train the MediaPipe's Gesture Recognition task with the chosen American Sign Language dataset and host it on IBM Cloud storage.
- Secondly, we connect the trained model with Dishari and start detecting and recognizing the hand gestures.
- Thirdly, the recognized hand gestures are fed as input to the search bar.
- Finally, we search the keyword generated by the hand gesture using Generative AI and generate the search results.

We describe each of the previously mentioned steps in detail in the following subsections. Initially, we start the next subsection by depicting system architecture(refer to Sect. 3.2.1).

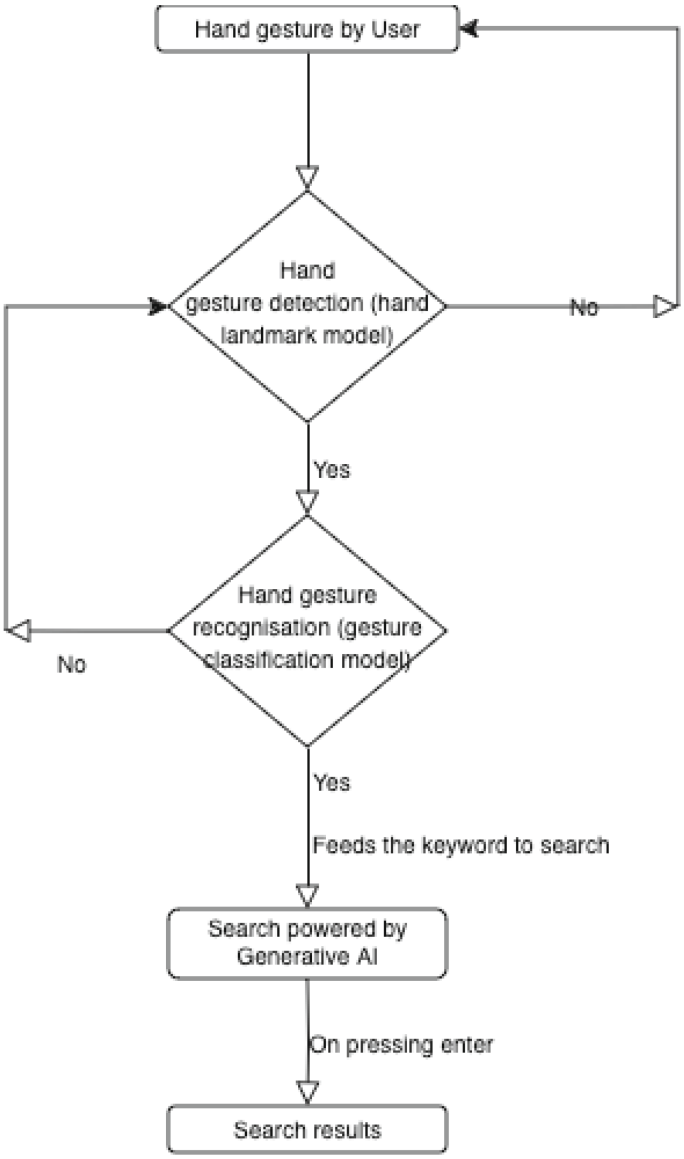


Fig. 2. Dishari flowchart

3.2.1 System Architecture

Dishari is a gesture-based web search application designed to facilitate hand gesture recognition for initiating keyword-based searches. The system architecture of Dishari involves several components working together seamlessly to achieve this functionality. At its core, Dishari utilizes Next.js which is a JavaScript framework for both the front-end user interface and the API for fetching search results.

Additionally, IBM Cloud is employed to store the trained dataset, providing scalability and reliability to the application. (refer Fig. 3)

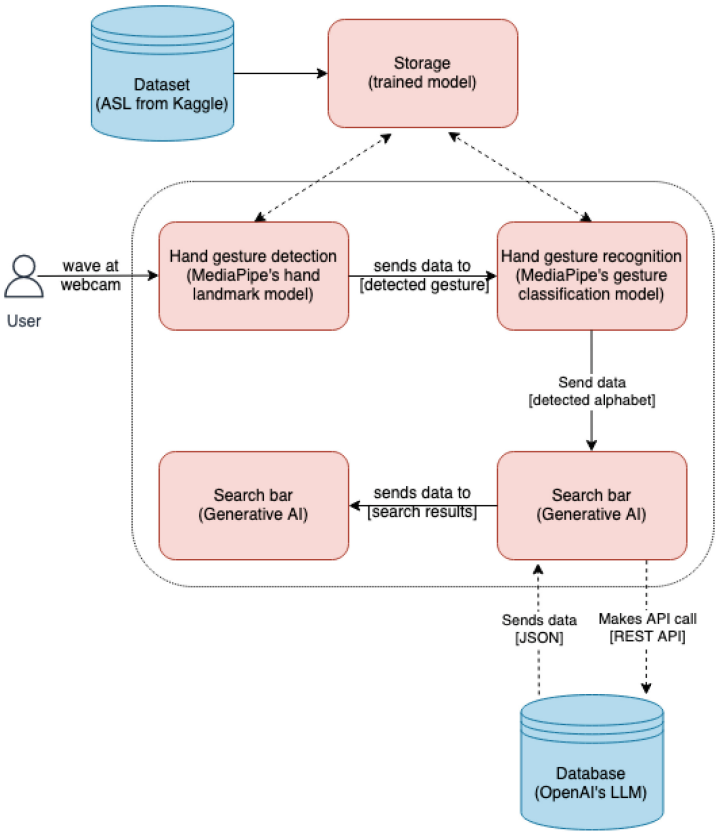


Fig. 3. Dishari system architecure

The front-end interface of Dishari presents users with a simple and intuitive design. Users initiate the gesture recognition process by pressing and holding the space bar key. Once activated, users are guided to perform specific hand gestures based on the provided instructions. Upon releasing the space bar key, Dishari captures the hand gesture and sends it to the back end for processing (refer Sect. 3.2.2).

On the back end, Dishari leverages MediaPipe, a popular open-source framework for building machine learning pipelines, particularly well-suited for tasks such as hand gesture recognition. MediaPipe incorporates Convolutional Neural Networks (CNNs) for efficient and accurate gesture recognition. These CNNs are trained on vast datasets to learn patterns and features essential for recognizing hand gestures accurately (refer Sect. 3.2.3).

3.2.2 Gesture Detection and Recognition Using MediaPipe

Initially, once we press the spacebar, the web camera automatically detects the gestures (ASL) and MediaPipe's Gesture Recognizer module serves as a cornerstone in Dishari's gesture recognition process. This module integrates two crucial pre-packaged model bundles: a hand landmark model bundle and a gesture classification model bundle. These bundles work in tandem to enable accurate and efficient recognition of hand gestures.

The hand landmark model bundle is responsible for detecting the presence of hands within the captured video frames and extracting essential geometric features. By analyzing the hand geometry, including landmarks such as fingertips, joints, and palm contours, this model accurately identifies the spatial configuration of the hand in real-time.

Simultaneously, the gesture classification model bundle leverages the hand geometry data provided by the landmark model to recognize specific gestures performed by the user. This model is trained on a diverse dataset encompassing various hand gestures, enabling it to classify incoming hand configurations with high accuracy.

During inference, the hand landmark model processes the incoming video frames, identifying and localizing the hands present in the frame. The extracted hand geometry information is then fed into the gesture classification model(CNN), which predicts the most likely gesture based on the learned patterns and features.

By utilizing these pre-trained model bundles, Dishari benefits from MediaPipe's robust and efficient gesture recognition capabilities. The integration of the hand landmark and gesture classification models enables Dishari to accurately interpret user hand gestures in real-time, facilitating seamless interaction with the application. We discuss how we train gesture recognition in Sect. 3.2.3.

3.2.3 Training the Gesture Recognition Model

Training the gesture recognition model within Dishari involves fine-tuning the pre-packaged gesture classification model bundle provided by MediaPipe. While the hand landmark model does not require training as it is pre-trained to detect hand geometry, the gesture classification model requires adaptation to specific gesture recognition tasks.

In parallel, the gesture classification model bundle operates on the extracted hand geometry data to recognize specific gestures performed by the user. This model is trained on a diverse dataset, including the American Sign Language dataset sourced from Kaggle. Through extensive training, the gesture classification model learns to associate distinctive hand gestures with corresponding semantic meanings, enabling accurate interpretation of user inputs.

The training dataset encompasses a wide range of hand gestures, providing ample variations and scenarios for training the model effectively. By iteratively optimizing model parameters and training configurations, Dishari ensures that the gesture classification model achieves high accuracy and robustness in recognizing user gestures.

Throughout the training phase, emphasis is placed on generalization and adaptability, enabling the model to perform reliably across diverse environmental conditions and user interactions. Data augmentation techniques may be employed to enhance the model's resilience to variations in lighting, background clutter, and hand orientation, further improving its real-world applicability.

By leveraging the hand landmark and gesture classification model bundles provided by MediaPipe and training them with the ASL dataset from Kaggle, Dishari achieves a sophisticated level of hand gesture recognition capability. This integration of advanced machine learning models with real-time computer vision techniques empowers Dishari to deliver a seamless and intuitive user experience for initiating keyword-based searches through hand gestures.

3.2.4 Connecting Hand Gesture Recognition with Search Powered by Generative AI

Dishari harnesses the power of Generative AI, particularly OpenAI's Generative Pre-trained Transformers (GPT), and Large Language Models (LLMs), for search result generation. Generative AI refers to a class of machine learning algorithms capable of generating new content, such as text or images, based on learned patterns and input prompts.

OpenAI's GPT models, including GPT-3, are among the most advanced LLMs available, pre-trained on vast amounts of text data to understand and generate human-like language. These models excel in tasks such as text completion, summarization, and question answering, making them ideal for generating search results based on user queries.

Within Dishari, the OpenAI completions API serves as the backbone for search result generation. When a user initiates a search query through a hand gesture, Dishari captures the query and sends it to the OpenAI API. The API then leverages the underlying GPT model to generate a list of search results tailored to the user's query.

The beauty of using Generative AI for search result generation lies in its ability to understand and contextualize natural language prompts, enabling it to produce relevant and coherent search results. By tapping into the vast knowledge encoded within the GPT model, Dishari provides users with insightful and informative search outcomes, enhancing their browsing experience.

Moreover, the integration of Generative AI opens up avenues for personalized and adaptive search experiences within Dishari. As users interact with the application and provide feedback on search results, the model can learn and adapt to user preferences over time, delivering increasingly tailored and relevant content.

In summary, by leveraging OpenAI's Generative AI capabilities, Dishari revolutionizes the search experience, offering users a seamless and intuitive way to explore and discover content. As Generative AI continues to evolve, the potential for enhancing search result generation within Dishari remains boundless, promising exciting prospects for the future of human-computer interaction.

We mention the pseudocode to implement Dishari. Please refer to the following pseudocode.

3.3 Pseudocode

This pseudocode outlines the main algorithm for the Dishari web application, including gesture recognition and search functionalities (refer Algorithm 1).

The Algorithm 1 (Dishari Gesture Recognition and Search) orchestrates the entire process of the Dishari web application, taking an input video stream as its parameter. It proceeds by first detecting the hand gesture present in the video using the MediaPipe's DetectGesture subroutine (refer Algorithm 2), then mapping this gesture to a corresponding keyword through MapGestureToKeyword. Subsequently, it utilizes the generated keyword to perform a search operation using the GenerativeSearch subroutine (refer Algorithm 3). Finally, the search results are displayed to the user via the application interface, providing them with relevant information based on the keyword which was formed from the recognized gesture.

Algorithm 1. Dishari Gesture Recognition and Search

```
procedure MAIN(input_video)
    gesture ← DETECTGESTURE(input_video)
    keyword ← MAPGESTURETOKEYWORD(gesture)
    search_results ← GENERATIVESEARCH(keyword)
    DISPLAYRESULTS(search_results)
end procedure
```

The Algorithm 2 (Detect Gesture) is responsible for identifying and classifying hand gestures present in the input video stream. It utilizes the MediaPipe library to extract hand landmarks from the video frames, which are then passed through a gesture classification model to determine the specific gesture being performed. The output of this subroutine is the recognized gesture, which is subsequently used for keyword mapping and search operations (refer Algorithm 2).

Algorithm 2. Detect Gesture

```
procedure DETECTGESTURE(input_video)
    hand_landmarks ← MEDIAPIPEHANDLANDMARKS(input_video)
    gesture ← CLASSIFYGESTURE(hand_landmarks)
    return gesture
end procedure
```

The Algorithm 3 (Generative Search) handles the process of retrieving search results based on a given keyword. It interacts with the application's search engine API, passing the keyword as input and receiving relevant search results in return. These results are then returned by the subroutine to be displayed to the user, completing the search functionality of the Dishari web application (refer Algorithm 3).

Algorithm 3. Generative Search

```
procedure GenerativeSearch(keyword)
    search_results ← SearchEngineAPI(keyword)
    return search_results
end procedure
```

This pseudocode provides a comprehensive overview of Dishari's operation, detailing the steps involved in gesture recognition, search query generation, and search functionality. By breaking down the application's processes into functional components, this pseudocode serves as a blueprint for implementing Dishari's functionality in a programming language of choice.

4 Results and Observations

We define the user demographics and their feedback in terms of evaluating Dishari in this section.

4.1 User Demographic

Dishari yields promising results based on both technical performance metrics and user feedback obtained through a survey conducted among 160 participants aged between 20 to 30. Refer to the below charts for the Demographics and Age of the people involved in the survey. (refer Figs. 4, 5).

Figure 4 represents the percentage of the race of the users that are involved in the feedback survey. While Asians are the highest with 95.1%, African Americans are the least with 0.6% and there are 4.3% of Other races.

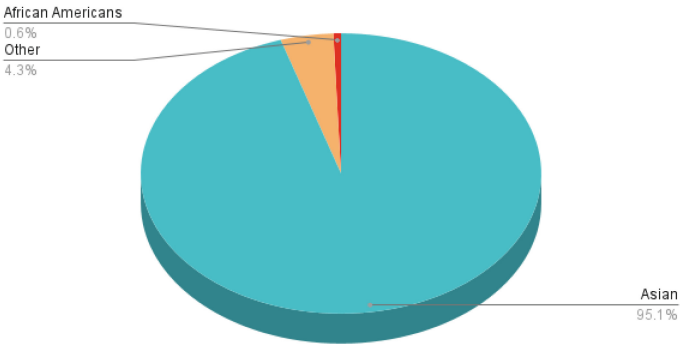


Fig. 4. Demographics

Figure 5 depicts the age of all the users that are involved in the feedback survey which is ranging majorly between 20 to 30.

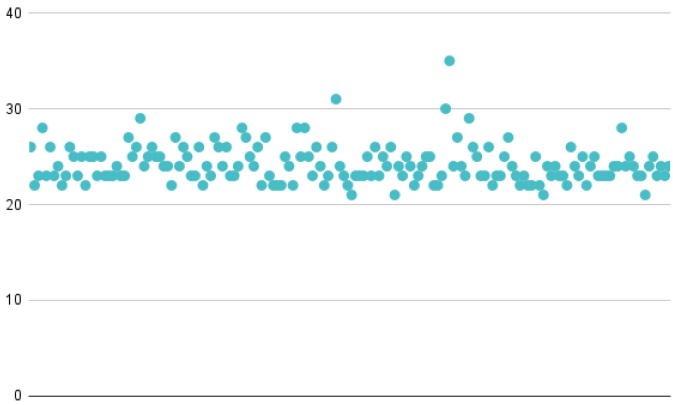


Fig. 5. Age

4.2 Technical Performance

The training of the gesture recognition model within Dishari yielded promising results, as evidenced by the training and validation metrics obtained during the training process. The model architecture, comprising a hand embedding layer, batch normalization, ReLU activation, dropout layer, and a custom gesture recognizer output layer, was trained over 10 epochs using a large dataset of hand gesture samples.

Throughout the training process, the model demonstrated steady improvement in both categorical accuracy and loss metrics. The **categorical accuracy steadily increased from approximately 82.85% during the first epoch to around 86.99% by the final epoch**. Conversely, the **loss metric gradually decreased from an initial value of 0.4893 to 0.3719, indicating that the model learned to better classify hand gestures over successive epochs** (refer Fig. 6).

```
Epoch 1/10
57278/57278 [==============================] - 403s 7ms/step - loss: 0.4893 - categorical_accuracy: 0.8285 - val_loss: 0.0402 - val_categorical_accuracy: 0.9395
Epoch 2/10
57278/57278 [==============================] - 406s 7ms/step - loss: 0.4018 - categorical_accuracy: 0.8597 - val_loss: 0.0858 - val_categorical_accuracy: 0.9300
Epoch 3/10
57278/57278 [==============================] - 391s 7ms/step - loss: 0.3929 - categorical_accuracy: 0.8621 - val_loss: 0.0598 - val_categorical_accuracy: 0.9354
Epoch 4/10
57278/57278 [==============================] - 399s 7ms/step - loss: 0.3851 - categorical_accuracy: 0.8659 - val_loss: 0.0660 - val_categorical_accuracy: 0.9345
Epoch 5/10
57278/57278 [==============================] - 403s 7ms/step - loss: 0.3827 - categorical_accuracy: 0.8657 - val_loss: 0.0652 - val_categorical_accuracy: 0.9365
Epoch 6/10
57278/57278 [==============================] - 412s 7ms/step - loss: 0.3793 - categorical_accuracy: 0.8672 - val_loss: 0.0946 - val_categorical_accuracy: 0.9328
Epoch 7/10
57278/57278 [==============================] - 412s 7ms/step - loss: 0.3784 - categorical_accuracy: 0.8678 - val_loss: 0.1072 - val_categorical_accuracy: 0.9237
Epoch 8/10
57278/57278 [==============================] - 396s 7ms/step - loss: 0.3757 - categorical_accuracy: 0.8691 - val_loss: 0.1110 - val_categorical_accuracy: 0.9279
Epoch 9/10
57278/57278 [==============================] - 401s 7ms/step - loss: 0.3736 - categorical_accuracy: 0.8685 - val_loss: 0.1229 - val_categorical_accuracy: 0.9221
Epoch 10/10
57278/57278 [==============================] - 399s 7ms/step - loss: 0.3719 - categorical_accuracy: 0.8699 - val_loss: 0.1010 - val_categorical_accuracy: 0.9235
```

Fig. 6. Epochs data during the training of the model

In the context of Dishari's model training results, the gradual decrease in loss metrics and increase in categorical accuracy over the 10 epochs suggest that the model is learning from the training data effectively. However, the slight disparity between the performance on the training and validation sets, particularly in the later epochs, could indicate a mild tendency towards overfitting.

Upon completion of training, the trained model was evaluated on a separate test dataset to assess its generalization performance. The test results revealed a **test loss of 0.0973 and a test accuracy of approximately 92.91%** (refer Fig. 7), indicating that the model performs well on unseen data, thereby validating its effectiveness in real-world scenarios.

Evaluate the model performance

After training the model, evaluate it on a test dataset and print the loss and accuracy metrics.

```
1 loss, acc = model.evaluate(test_data, batch_size=1)
2 print(f"Test loss:{loss}, Test accuracy:{acc}")
```

```
14320/14320 [==============================] - 112s 3ms/step - loss: 0.0973 - categorical_accuracy: 0.9291
Test loss:0.09727732837200165, Test accuracy:0.9291201233863831
```

Fig. 7. Model's Loss and Accuracy metrics after evaluating with test data

4.3 User Feedback

The survey results indicated high levels of satisfaction among users regarding various aspects of Dishari's usability and user interface:

- **Convenience of Using Hand Gestures:** Participants rated the convenience of using hand gestures in Dishari very positively, with an average score of 4.53 out of 5 (refer Fig. 8). This suggests that users found hand gestures to be a convenient and intuitive input method for initiating searches.
  Figure 8 depicts the ratings convenience of using hand gestures in Dishari from those involved in the user feedback survey. On the x-axis, numbers are listed as per the rating scale, i.e., 1 to 5. The y-axis shows the total number of users who rated that particular number.

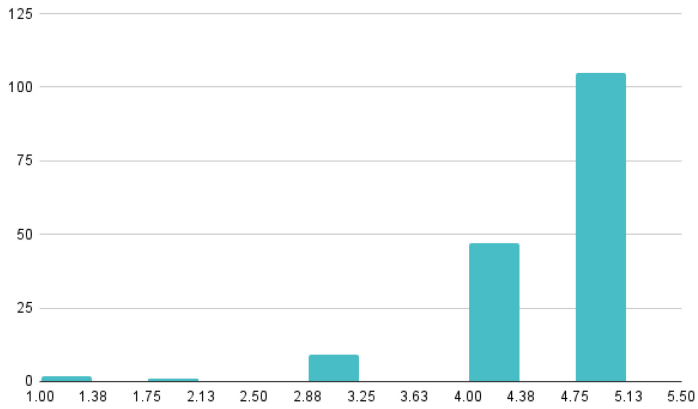


Fig. 8. Convenience of using hand gestures rating

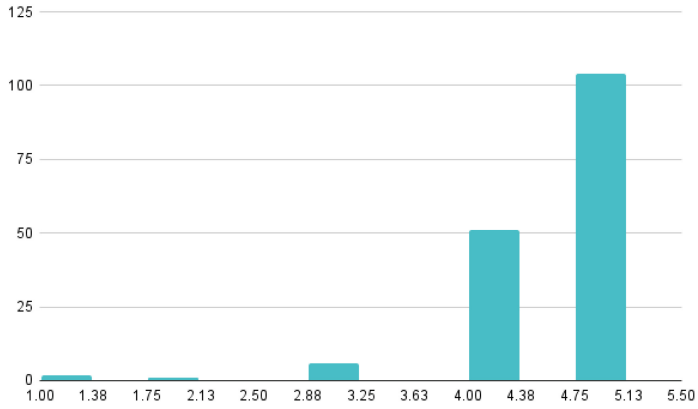


Fig. 9. Ease of use rating

- **Ease of Use:** Dishari received high ratings for ease of use, with an average score of 4.54 out of 5 (refer Fig. 9). This indicates that users found the application easy to navigate and interact with, contributing to a positive user experience.
  Figure 9 depicts the ratings of the ease of use of Dishari from those involved in the user feedback survey. On the x-axis, numbers are listed as per the rating scale, i.e., 1 to 5. The y-axis shows the total number of users who rated that particular number.
- **User Interface (UI) Ratings:** Participants rated various aspects of Dishari's UI highly, with average scores ranging from 4.58 to 4.67 out of 5 (refer Fig. 10). Specifically, the webcam interface, instructions for using

Dishari, search bar, search results display, and hand gesture guide all received favorable ratings. These scores reflect the effectiveness of Dishari's UI design in guiding users through the gesture recognition process and facilitating smooth interactions.
Fig. 10 depicts the average of all the ratings of different User Interface (UI) sections of Dishari from those involved in the user feedback survey. The sections listed on the x-axis are UI-Webcam, UI-Instructions to use Dishari, UI-Search bar, and UI-Hand gestures. The y-axis shows the scale of the rating (1 to 5).

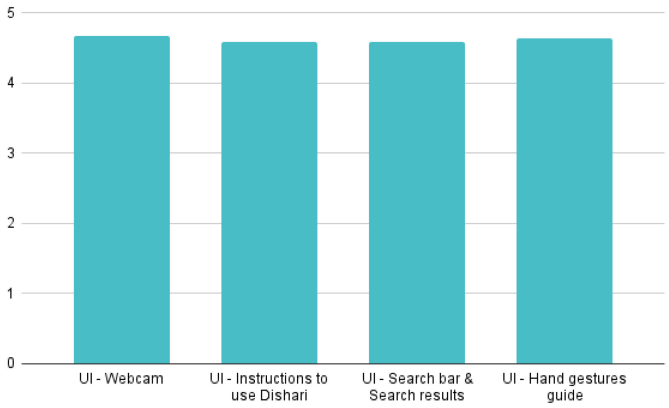


Fig. 10. UI ratings

- **User Experience (UX) Ratings:** Users also provided positive feedback on the interaction aspects of Dishari, with average scores ranging from 4.52 to 4.65 out of 5 (refer Fig. 11). This includes ratings for hand gesture detection, recognition, search bar interaction, and the hand gesture guide. The high scores indicate that users found the interaction with Dishari to be intuitive and responsive, enhancing their overall satisfaction with the application.
Fig. 11 depicts the average of all the ratings of different User Experience (UX) sections of Dishari from those involved in the user feedback survey. The sections listed on the x-axis are UX-Hand gesture detection, UX-Hand gesture recognition, UX-Search bar & Search results, and UX-Hand gestures guide. The y-axis shows the scale of the rating (1 to 5).

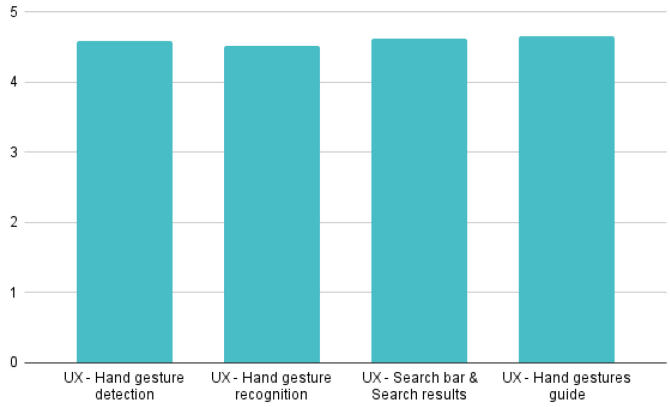


Fig. 11. UX ratings

The results and observations indicate that Dishari effectively bridges the gap between gesture-based interaction and keyword-based search, offering users a convenient and intuitive way to interact with web applications. The high satisfaction levels among users, coupled with the robust performance of the gesture recognition model, underscore the potential of Dishari to enhance user experience and accessibility in digital environments. Moving forward, further refinement and optimization of the application based on user feedback will contribute to its continued success and adoption.

5 Conclusion and Future Scope

In conclusion, Dishari represents a significant advancement in the field of human-computer interaction, leveraging state-of-the-art technologies to enable intuitive gesture-based interactions for initiating keyword-based searches. The integration of MediaPipe's Gesture Recognizer module, coupled with generative AI-powered search, provides users with a novel and efficient means of interacting with web applications.

Through the implementation of Dishari, we have demonstrated the feasibility and effectiveness of using hand gestures as input for initiating search queries, thereby enhancing accessibility and usability for specially-abled people. The robustness and accuracy of the gesture recognition system, coupled with the personalized search results generated by generative AI, contribute to an enhanced user experience.

Looking ahead, there are several avenues for further research and development to enhance the capabilities and functionalities of Dishari. One potential direction is the expansion of gesture vocabulary to encompass a broader range of gestures, including those specific to different sign languages or cultural contexts.

Furthermore, exploring the integration of additional modalities such as facial expressions or body movements could enrich the gesture recognition system, enabling more nuanced and contextually aware interactions. Additionally, ongoing advancements in machine learning and computer vision algorithms may lead to further improvements in gesture recognition accuracy and real-time performance.

Overall, Dishari lays the foundation for future research and innovation in the domain of gesture-based interaction systems, with the potential to revolutionize how users interact with web applications and digital interfaces in the years to come.

References

1. Bhattacharya, T., Qin, X.: Modeling energy efficiency of future green data centers. In: 2020 11th International Green and Sustainable Computing Workshops (IGSC), pp. 1–3. IEEE (2020)
2. Bhattacharya, T., et al.: Capping carbon emission from green data centers. Int. J. Energy Environ. Eng. **14**(4), 627–641 (2023)
3. Mooers, C.N.: Application of random codes to the gathering of statistical information. Ph.D. dissertation, Massachusetts Institute of Technology (1948)
4. El Guemmat, K., Ouahabi, S.: A literature review of indexing and searching techniques implementation in educational search engines. Int. J. Inf. Commun. Technol. Educ. (IJICTE) **14**(2), 72–83 (2018)
5. Guemmat, K.E., Talea, M., et al.: Implementation and evaluation of an indexing model of teaching and learning resources. Procedia. Soc. Behav. Sci. **191**, 1266–1274 (2015)
6. Biletskiy, Y., Baghi, H., Steele, J., Vovk, R.: A rule-based system for hybrid search and delivery of learning objects to learners. Interact. Technol. Smart Educ. **9**(4), 263–279 (2012)
7. Ahmed-Ouamer, R., Hammache, A.: Ontology-based information retrieval for e-learning of computer science. In: 2010 International Conference on Machine and Web Intelligence, pp. 250–257. IEEE (2010)
8. Galvano, F.: Beyond borders: an in-depth analysis of cultural variances in non-verbal communication through gestures and hands (1969)
9. Gota, D.-I., Puscasiu, A., Fanca, A., Valean, H., Miclea, L.: Human-computer interaction using hand gestures. In: 2020 IEEE International Conference on Automation, Quality and Testing, Robotics (AQTR), pp. 1–5. IEEE (2020)
10. Guarino, A., Malandrino, D., Zaccagnino, R., Capo, C., Lettieri, N.: Touchscreen gestures as images. a transfer learning approach for soft biometric traits recognition. Expert Syst. Appl. **219**, 119614 (2023)
11. Núñez-Marcos, A., Perez-de Viñaspre, O., Labaka, G.: A survey on sign language machine translation. Expert Syst. Appl. **213**, 118993 (2023)
12. Obi, Y., Claudio, K.S., Budiman, V.M., Achmad, S., Kurniawan, A.: Sign language recognition system for communicating to people with disabilities. Procedia Comput. Sci. **216**, 13–20 (2023)
13. Sau, D.: ASL(American Sign Language) Alphabet Dataset (2021). Accessed 13 Oct 2021. https://www.kaggle.com/datasets/debashishsau/aslamerican-sign-language-aplhabet-dataset

14. Chatterjee, K., et al.: Hack: Hand gesture classification using a convolutional neural network and generative adversarial network-based data generation model. Information **15**(2), 85 (2024)
15. Rane, N.: Chatgpt and similar generative artificial intelligence (ai) for smart industry: role, challenges and opportunities for industry 4.0, industry 5.0 and society 5.0. Chall. Opport. Ind. **4** (2023)
16. Cao, Y., et al.: A comprehensive survey of ai-generated content (aigc): a history of generative ai from gan to chatgpt. arXiv preprint arXiv:2303.04226 (2023)
17. Oppenlaender, J.: A taxonomy of prompt modifiers for text-to-image generation. Behav. Inf. Technol., 1–14 (2023)
18. Ooi, K.-B., et al.: The potential of generative artificial intelligence across disciplines: Perspectives and future directions. J. Comput. Inf. Syst., 1–32 (2023)
19. Kshetri, N.: Generative artificial intelligence and e-commerce. Computer **57**(2), 125–128 (2024)
20. Liu, G., et al.: Semantic communications for artificial intelligence generated content (aigc) toward effective content creation. IEEE Netw. (2024)
21. Zhang, P., Boulos, M.: Generative ai in medicine and healthcare: promises, opportunities and challenges. Future Internet **15**(9), 286 (2023)
22. Ayoola, O.O., Alenoghena, R., Adeniji, S.: Chatgpt impacts on access-efficiency, employment, education and ethics: the socio-economics of an AI language model. BizEcons Q. **16**, 1–17 (2023)
23. Natarajan, B., et al.: Development of an end-to-end deep learning framework for sign language recognition, translation, and video generation. IEEE Access **10**, 104358–104374 (2022)
24. Kuzlu, M., Xiao, Z., Sarp, S., Catak, F.O., Gurler, N.., Guler, O.: The rise of generative artificial intelligence in healthcare. In: 2023 12th Mediterranean Conference on Embedded Computing (MECO), pp. 1–4. IEEE (2023)
25. Iqbal, J., et al.: Reimagining healthcare: unleashing the power of artificial intelligence in medicine. Cureus **15**(9), 1–14 (2023)
26. Harvey, E., Sandhaus, H., Jacobs, A.Z., Moss, E., Sloane, M.: The cadaver in the machine: The social practices of measurement and validation in motion capture technology. arXiv preprint arXiv:2401.10877 (2024)
27. Eid, A., Schwenker, F.: Visual static hand gesture recognition using convolutional neural network. Algorithms **16**(8), 361 (2023)
28. Bakheet, S., Al-Hamadi, A.: Robust hand gesture recognition using multiple shape-oriented visual cues. EURASIP J. Image Video Process. **2021**(1), 1–18 (2021). https://doi.org/10.1186/s13640-021-00567-1
29. Rong, Y., Gu, G.: Deep transfer learning-based adaptive gesture recognition of a soft e-skin patch with reduced training data and time. Sens. Actuators, A **363**, 114693 (2023)
30. Escalera, S., Athitsos, V., Guyon, I.: Challenges in multi-modal gesture recognition. In: Gesture Recognition, pp. 1–60 (2017)
31. Oudah, M., Al-Naji, A., Chahl, J.: Hand gesture recognition based on computer vision: a review of techniques. J. Imaging **6**(8), 73 (2020)
32. Shuvo, M., Islam, S.K., Cheng, J., Morshed, B.I.: Efficient acceleration of deep learning inference on resource-constrained edge devices: a review. Proc. IEEE **111**, 42–91 (2022)
33. Vatavu, R.-D.: Gesture-based interaction. In: Handbook of Human Computer Interaction, pp. 1–47. Springer, Heidelberg (2023)
34. Fiorini, L., et al.: Daily gesture recognition during human-robot interaction combining vision and wearable systems. IEEE Sensors J. **21**(20), 23568–23577 (2021)

35. Wenzel, M.: Generative adversarial networks and other generative models. In: Machine Learning for Brain Disorders, pp. 139–192 (2023)
36. Sabry, E.S., et al.: Matching evaluation based on image content discriminative features for different image types. Imaging Sci. J. **72**(1), 23–51 (2024)
37. Koh, J.Y., Baldridge, J., Lee, H., Yang, Y.: Text-to-image generation grounded by fine-grained user attention. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pp. 237–246 (2021)
38. Li, J., Tang, T., Zhao, W.X., Nie, J.-Y., Wen, J.-R.: Pretrained language models for text generation: a survey. arXiv preprint arXiv:2201.05273 (2022)
39. Wach, K., et al.: The dark side of generative artificial intelligence: a critical analysis of controversies and risks of chatgpt. Entrepreneurial Bus. Econ. Rev. **11**(2), 7–30 (2023)
40. Zhang, Y., et al.: Why does little robustness help? a further step towards understanding adversarial transferability. In: Proceedings of the 45th IEEE Symposium on Security and Privacy (S&P'24), vol. 2 (2024)
41. Lim, W., Chek, K., Theng, L.B., Lin, C.: Future of generative adversarial networks (gan) for anomaly detection in network security: a review. Comput. Secur. **139**, 103733 (2024)